Pooling and spatial information

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Pooling and spatial information before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Convolution as a local operation

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Learned representations

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Regularisation and early stopping

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

A fictional single-channel feature map has rows [1, 4, 2, 0], [3, 2, 5, 1], [0, 6, 1, 3] and [2, 4, 2, 7]. Use a 2 by 2 window, stride 2, no padding and only complete windows. Coordinates are zero-based.

The four starts are [0, 0], [0, 2], [2, 0] and [2, 2].

Max pooling keeps the largest value in each window. **Mean pooling** adds the four values and divides by 4. Neither operation learns weights in this lesson.

These are fixed summaries of the incoming feature map, unlike the trainable kernel parameters introduced last week.

Window start	Values	Maximum	Mean
[0, 0]	1, 4, 3, 2	4	2.5
[0, 2]	2, 0, 5, 1	5	2
[2, 0]	0, 6, 2, 4	6	3
[2, 2]	1, 3, 2, 7	7	3.25

The max **output** is [[4, 5], [6, 7]], and the mean **output** is [[2.5, 2], [3, 3.25]]. Both have shape 2 by 2. Sixteen values become four, but the four **output**s still preserve the coarse arrangement of the four regions.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Pooling and spatial information trace and record the first step you can explain without help.

Use W2: Prove that location can be lost for Pooling and **spatial information**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Pooling and spatial information explanation against the supplied facts before discussing it.

Use the worked case for Pooling and **spatial information**. Proposed explanation: “**Max pooling** turns negative numbers into zero”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Pool a new map independently

Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map. Explain why replacing a negative maximum by zero is an error here.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Calculate both summaries

> Calculate both summaries. Apply the supplied Pooling and spatial information facts and keep your reasoning.

For the lesson window starting at [2, 2], list the four values, sum, maximum and mean. State the complete pooled **output** shape.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Prove that location can be lost

> Prove that location can be lost. Apply the supplied Pooling and spatial information facts and keep your reasoning.

Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

> **My reasoning and evidence**

WORKBOOK · 7 OF 15

W3 Audit an omitted border

> Audit an omitted border. Apply the supplied Pooling and spatial information facts and keep your reasoning.

Apply the lesson complete-window rule to [[1, 2, 99], [3, 4, 99], [99, 99, 99]]. Give the **output** shape and both **output**s. Explain why neither **output** includes 99.

> **My reasoning and evidence**

WORKBOOK · 8 OF 15

W4 Pool a new map independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Pool a new map independently. Apply the supplied Pooling and spatial information facts and keep your reasoning.

Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map. Explain why replacing a negative maximum by zero is an error here.

My reasoning and evidence

W5 Check the movement boundary

> Check the movement boundary. Apply the supplied Pooling and spatial information facts and keep your reasoning.

A 2 by 4 map is zero except for 8 at row 0 column 1. Move the 8 to column 2 and set its old position to zero. Give the max and mean maps before and after. What claim does this refute?

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Pooling and spatial information to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled pooling and **spatial information evidence** sheet to Image-pattern Network Investigation. Show how you can compute max and **mean pooling** under an explicit window rule.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Pooling and spatial information

> Pool a small map and show the detail it cannot retain. Explain the labelled step before checking the changed condition.

Pool a small map and show the detail it cannot retain.

Feature map rows [1, 4, 2, 0],[3, 2, 5, 1]. Pool non-overlapping 2×2 windows, stride 2, no padding.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/22

Two adjacent pooling windows

	left 0	left 1	right 0	right 1
row 0	1	4	2	0
row 1	3	2	5	1

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/22

Pooling and spatial information

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Left window

Values 1, 4, 3, 2: max 4, mean 2.5.



2. Right window

Values 2, 0, 5, 1: max 5, mean 2.



3. Compare summaries

Max map [[4, 5]], mean map [[2.5, 2]], both shape 1 x 2.

CHANGE ONE THING

Swap 1 and 4 within the left window.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week’s input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Pool a small map and show the detail it cannot retain.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2.

1. Left window

2. Compare summaries

3. Right window

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare summaries Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2. Pooling compresses each local window into a summary.

Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Swap 1 and 4 within the left window. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Both pooled maps remain unchanged.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Both pooled maps remain unchanged. Max and mean ignore the positions of values inside the same window.

Repair a missing building block

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learned representations (ai-5/19). Goal: Trace an **input** through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regularisation and early stopping (ai-5/16). Goal: Separate a weight penalty from **prediction** error. Short practice: Why should the penalty value not be reported as validation error?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

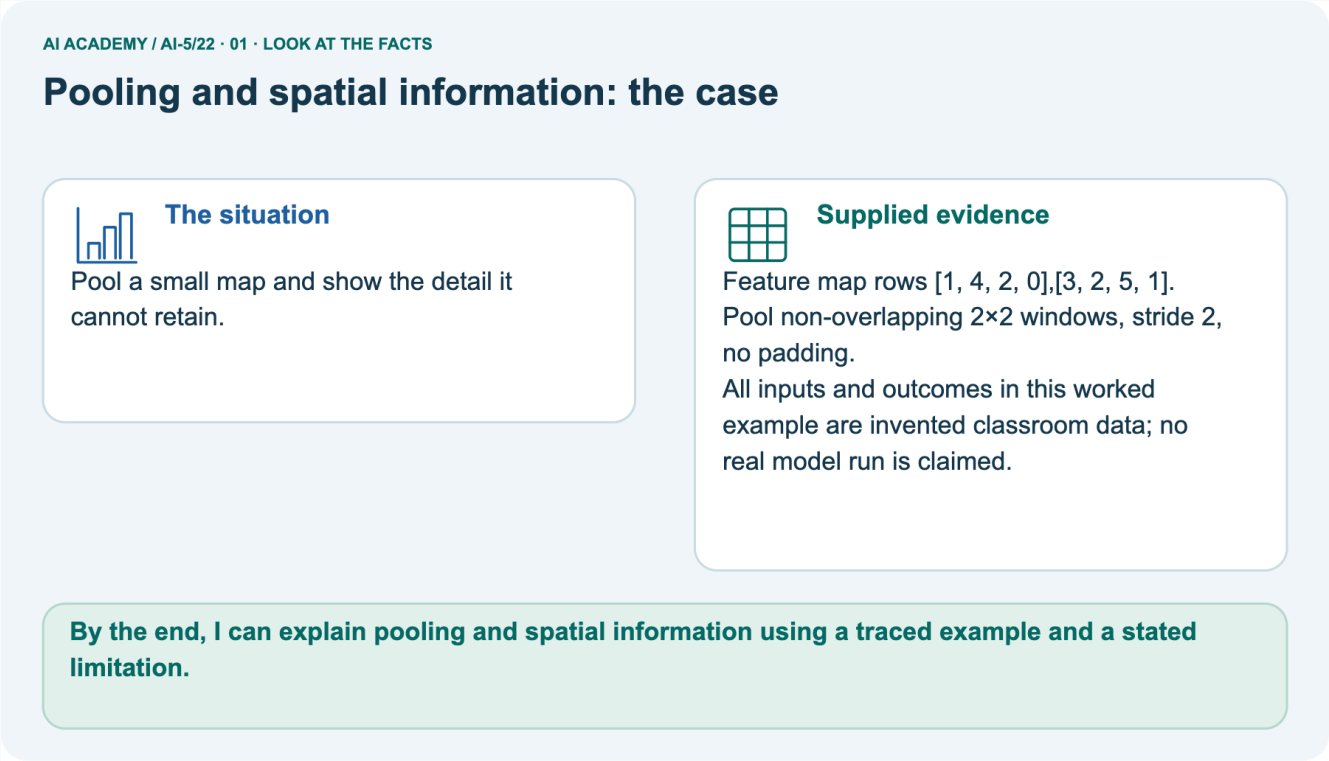
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Pooling and spatial information picture from its labels, then record a relevant limit.

01. Pooling and spatial information: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

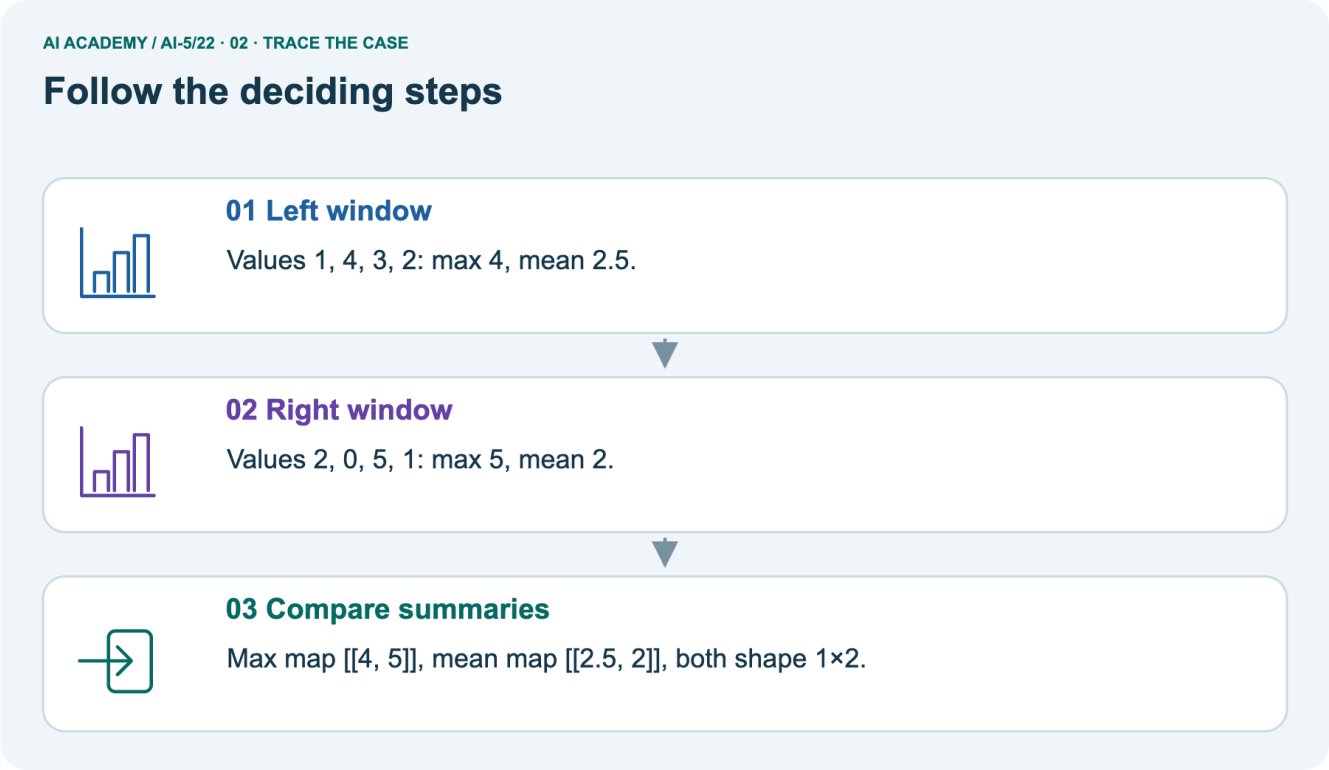
Pool a small map and show the detail it cannot retain. Supplied facts: Feature map rows [1, 4, 2, 0], [3, 2, 5, 1]. Pool non-overlapping 2×2 windows, stride 2, no padding.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Left window: Values 1, 4, 3, 2: max 4, mean 2.5. → Right window: Values 2, 0, 5, 1: max 5, mean 2. → Compare summaries: Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two adjacent pooling windows



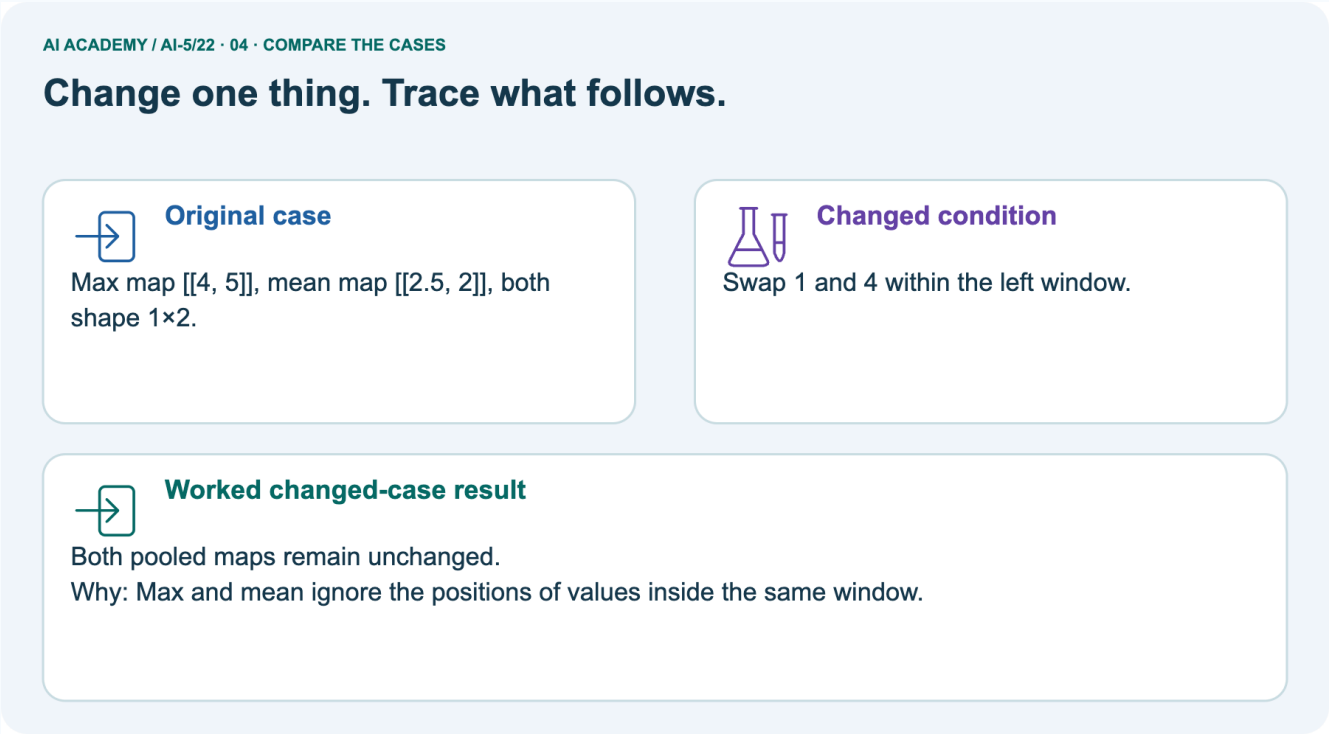
Follow the labelled picture with the matching explanation. Open the vector diagram

Two adjacent pooling windows. row 0: left 0 = 1, left 1 = 4, right 0 = 2, right 1 = 0. row 1: left 0 = 3, left 1 = 2, right 0 = 5, right 1 = 1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

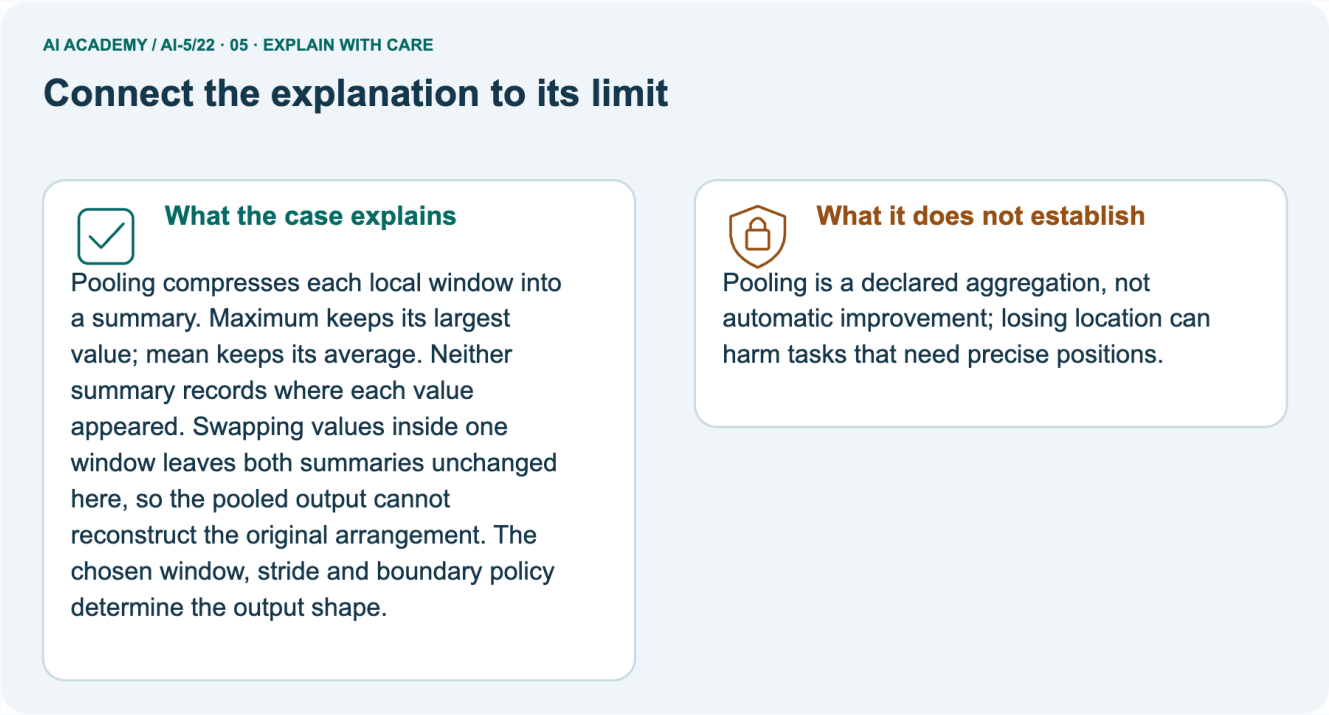
Original case: Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2. Changed condition: Swap 1 and 4 within the left window. Expected result: Both pooled maps remain unchanged.

Reason: Max and mean ignore the positions of values inside the same window.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Pooling compresses each local window into a summary. Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

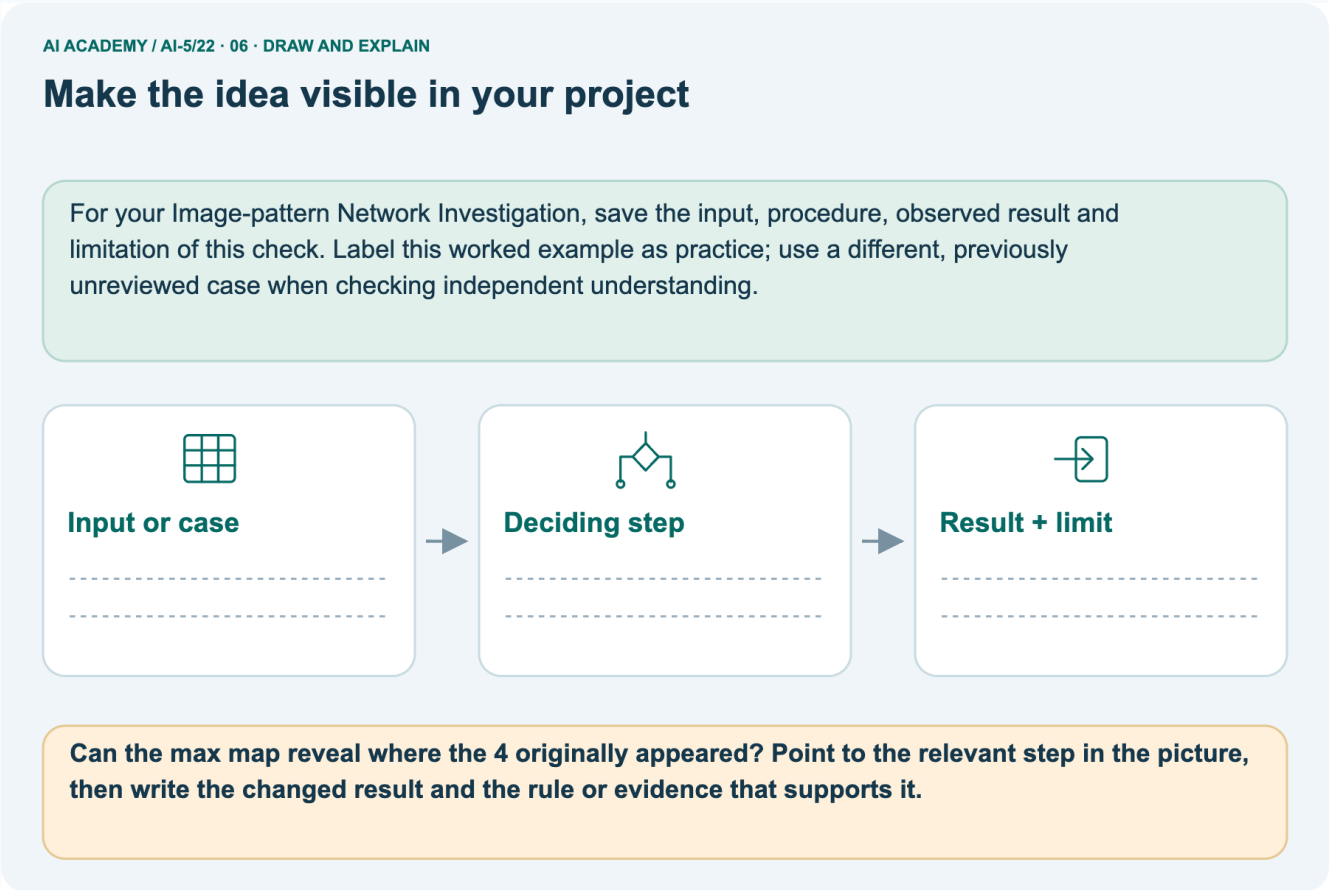
Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

Limit: Pooling is a declared aggregation, not automatic improvement; losing location can harm tasks that need precise positions.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.